MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

61. The area of the region

$$A=\left\{(x,\ y):|\cos x-\sin x|\le y\le \sin x,\ 0\le x\le \frac{\pi}{2}\right\}$$

is

(1) $\sqrt{5}-2\sqrt{2}+1$

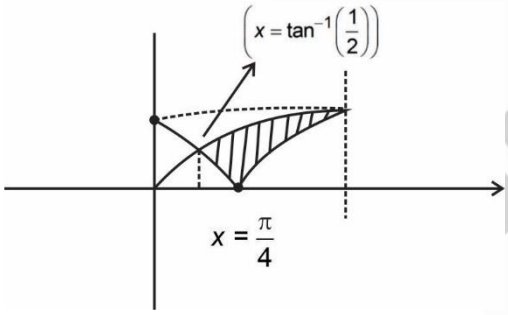
(2) $\frac{3}{\sqrt{5}}-\frac{3}{\sqrt{2}}+1$

(3) $\sqrt{5}+2\sqrt{2}-4.5$

(4) $1-\frac{3}{\sqrt{2}}+\frac{4}{\sqrt{5}}$

Answer (1)

Sol.



$$\text{Area}=\int_{\tan^{-1}\frac{1}{2}}^{\frac{\pi}{4}}\left(\sin x-(\cos x-\sin x)\right)dx+\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\left(\sin x-(\sin x-\cos x)\right)dx$$

$$=\int_{\tan^{-1}\frac{1}{2}}^{\frac{\pi}{4}}(2\sin x-\cos x)\,dx+(\sin x)_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$=2\cos x-\sin x\Big|_{\tan^{-1}\frac{1}{2}}^{\frac{\pi}{4}}+\left(1-\frac{1}{\sqrt{2}}\right)$$

$$=\frac{3}{\sqrt{2}}+\frac{1}{\sqrt{5}}+\frac{1}{\sqrt{5}}+1-\frac{1}{\sqrt{2}}$$

$$=\sqrt{5}-2\sqrt{2}+1$$

62. Let K be the sum of the coefficients of the odd powers of x in the expansion of $(1+x)^{99}$. Let a be the middle term in the expansion of $\left(2+\frac{1}{\sqrt{2}}\right)^{200}$. If $\frac{{}^{200}C_{99}K}{a}=\frac{2^l m}{n}$, where m and n are odd numbers, then the ordered pair (l, n) is equal to

(1) (51, 101) (2) (51, 99)

(3) (50, 101) (4) (50, 51)

Answer (3)

Sol. $K=2^{98}$

$a={}^{200}C_{100}\ 2^{50}$

$$\therefore\quad \frac{{}^{200}C_{99}\cdot 2^{98}}{{}^{200}C_{100}\cdot 2^{50}}=\frac{2^l m}{n}$$

$$\Rightarrow \frac{100}{101}\cdot 2^{48}=\frac{2^l m}{n}$$

$$\Rightarrow \frac{25}{101}\cdot 2^{50}=\frac{2^l m}{n}$$

$\therefore\ l=50,\ m=25,\ n=101$

63. If the lines $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z+3}{1}$ and $\frac{x-a}{2}=\frac{y+2}{3}=\frac{z-3}{1}$ intersect at the point P, then the distance of the point P from the plane $z=a$ is

(1) 28 (2) 16

(3) 10 (4) 22

Answer (1)

Sol. $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z+3}{1}=\lambda$ (say)

& $\frac{x-a}{2}=\frac{y+2}{3}=\frac{z-3}{1}=\mu$ (say)

$\therefore\ \lambda+1=2\mu+a$ ...(i)

$2\lambda+2=3\mu+2$ ...(ii)

$\lambda-3=\mu+3$ ...(iii)

By (i) & (ii)

$\Rightarrow\ 3\mu-2=4\mu+2a+2$

$\Rightarrow$ $\mu = -2(1 + a)$ & $\lambda = 5 - 3a$

Put λ & μ in (iii) we get

$a = -9$

$\mu = 16$

$\lambda = 22$

$\therefore$ Point of intersection $\equiv (23, 46, 19)$

Distance from z = – 9 is 28

64. If the tangent at a point P on the parabola $y^2 = 3x$ is parallel to the line $x + 2y = 1$ and the tangents at the points Q and R on the ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$ are perpendicular to the line $x - y = 2$, then the area of the triangle PQR is

(1) $5\sqrt{3}$ (2) $3\sqrt{5}$

(3) $\frac{9}{\sqrt{5}}$ (4) $\frac{3}{2}\sqrt{5}$

Answer (2)

Sol. $P \equiv \left(\frac{A}{m^2}, \frac{2A}{m}\right)$ where $\left(A = \frac{3}{4}, m = \frac{-1}{2}\right)$

& $Q, R = \left(\mp\frac{a^2 m_1}{a^2 m_1^2 + b^2}, \frac{\mp \cdot b^2}{\sqrt{a^2 m_1^2 + b^2}}\right)$

Where $a^2 = 4$, $b^2 = 1$ and $m_1 = 1$

$\therefore$ $P \equiv (3, -3)$

$Q \equiv \left(\frac{-4}{\sqrt{5}}, \frac{-1}{\sqrt{5}}\right)$ & $R\left(\frac{4}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$

$$\text{Area} = \frac{1}{2}\begin{vmatrix} 3 & -3 & 1 \\ \frac{-4}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & 1 \\ \frac{4}{\sqrt{5}} & \frac{1}{\sqrt{5}} & 1 \end{vmatrix} = \frac{1}{10}\begin{vmatrix} 3 & -3 & 1 \\ -4 & -1 & \sqrt{5} \\ 0 & 0 & 2\sqrt{5} \end{vmatrix}$$

$$= \frac{2\sqrt{5}}{10}(-15) = 3\sqrt{5}$$

65. The statement $B \Rightarrow ((\sim A) \vee B)$ is equivalent to

(1) $A \Rightarrow (A \Leftrightarrow B)$

(2) $B \Rightarrow ((\sim A) \Rightarrow B)$

(3) $B \Rightarrow (A \Rightarrow B)$

(4) $A \Rightarrow ((\sim A) \Rightarrow B)$

Answer (2, 3, 4)

Sol. $B \Rightarrow ((\sim A) \vee B) \equiv \sim B \vee (\sim A \vee B)$

$\equiv (\sim B \vee B) \vee (\sim A) \equiv T$

Option (2) $B \Rightarrow ((\sim A) \Rightarrow B) \equiv (\sim B) \vee (\sim A \Rightarrow B)$

$\equiv (\sim B) \vee (A \vee B) \equiv T$

Option (3) $B \Rightarrow (A \Rightarrow B) \equiv B \Rightarrow ((\sim A) \vee B)$

(same as given)

Option (4) $A \Rightarrow ((\sim A) \Rightarrow B) \equiv (\sim A) \vee (A \vee B) \equiv T$

66. The set of all value of $t \in R$, for which the matrix

$$\begin{bmatrix} e^t & e^{-t}(\sin t - 2\cos t) & e^{-1}(-2\sin t - \cos t) \\ e^t & e^{-t}(2\sin t + \cos t) & e^{-1}(\sin t - 2\cos t) \\ e^t & e^{-1}\cos t & e^{-1}\sin t \end{bmatrix}$$ is

invertible, is

(1) $\left\{(2k+1)\frac{\pi}{2},\ k \in Z\right\}$ (2) R

(3) $\left\{k\pi + \frac{\pi}{4},\ k \in Z\right\}$ (4) $\{k\pi,\ k \in Z\}$

Answer (2)

Sol. $$\begin{vmatrix} 0 & e^{-t}(-\sin t - 3\cos t) & e^{-t}(-3\sin t + \cos t) \\ 8 & e^{-t}(2\sin t) & e^{-t}(-2\cos t) \\ e^t & e^{-t}\cos t & e^{-t}\sin t \end{vmatrix} = 0$$

(If matrix is non invertible)

$-2\cos t(-\sin t - 3\cos t) - 2\sin t(\cos t - 3\sin t) = 0$

$\Rightarrow$ $6\cos^2 t + 6\sin^2 t = 0$

$t \in 0$

67. Let $S = \{w_1, w_2, ...\}$ be the sample space associated to a random experiment. Let $P(w_n) = \frac{P(w_{n-1})}{2}$, $n \geq 2$. Let $A = \{2k + 3l : k, l \in N\}$ and $B = \{w_n : n \in A\}$. Then $P(B)$ is equal to

(1) $\frac{1}{16}$ (2) $\frac{1}{32}$

(3) $\frac{3}{64}$ (4) $\frac{3}{32}$

Answer (3)

Sol. $P(w_1) + \frac{P(w_1)}{2} + \frac{P(w_1)}{2^2} + ... = 1$

$\therefore$ $P(w_1) = \frac{1}{2}$

Hence, $P(w_n)=\frac{1}{2^n}$

Every number except 1, 2, 3, 4, 6 is representable in the form

$2k+3l$ where $k,\ l \in N$.

$\therefore \quad P(B)=1-P(w_1)-P(w_2)$

$-P(w_3)-P(w_4)-P(w_6)$

$=\frac{3}{64}$

68. The value of the integral $\int_1^2\left(\frac{t^4+1}{t^6+1}\right)dt$ is

(1) $\tan^{-1}2-\frac{1}{3}\tan^{-1}8+\frac{\pi}{3}$

(2) $\tan^{-1}2+\frac{1}{3}\tan^{-1}8-\frac{\pi}{3}$

(3) $\tan^{-1}\frac{1}{2}+\frac{1}{3}\tan^{-1}8-\frac{\pi}{3}$

(4) $\tan^{-1}\frac{1}{2}-\frac{1}{3}\tan^{-1}8+\frac{\pi}{3}$

Answer (2)

Sol. $\int_1^2\frac{t^4+1}{t^6+1}dt$

$=\int_1^2\frac{\left(t^2+1\right)^2}{t^6+1}dt-2\int_1^2\frac{t^2}{t^6+1}dt$

$=\int_1^2\frac{t^2+1}{t^4-t^2+1}dt-2\int_1^2\frac{t^2}{\left(t^3\right)^2+1}dt$

$=\tan^{-1}\left(2t+\sqrt{3}\right)+\tan^{-1}\left(2t-\sqrt{3}\right)\Big|_1^2$

$-\frac{2}{3}\tan^{-1}\left(t^3\right)\Big|_1^2$

$=\tan^{-1}\left(4+\sqrt{3}\right)+\tan^{-1}\left(4-\sqrt{3}\right)-\tan^{-1}\left(2+\sqrt{3}\right)$

$-\tan^{-1}\left(2+\sqrt{3}\right)$

$-\tan^{-1}\left(2\sqrt{3}\right)-\frac{2}{3}\left(\tan^{-1}8-\tan^{-1}1\right)$

$=\tan^{-1}2+\frac{1}{3}\tan^{-1}8-\frac{\pi}{3}$

69. Consider a function $f:N\to R$, satisfying

$f(1)+2f(2)+3f(3)+....+xf(x)=x(x+1)f(x);x\geq 2$

with $f(1) = 1$. Then $\frac{1}{f(2022)}+\frac{1}{f(2028)}$ is equal to

(1) 8200 (2) 8400

(3) 8100 (4) 8000

Answer (3)

Sol. $f(1)+2f(2)+3f(3)+...+nf(n)=n(n+1)+(n)\ldots(i)$

$n\to n+1$

$f(1)+2f(2)+...+(n+1)f(n+1)=(n+1)(n+2)$

$f(n+1)$ ...(ii)

(i) and (ii) gives

$3f(3)-2f(2)=0$

$4f(4)-3f(3)=0$

$\vdots$

$(n+1)f(n+1)-nf(n)=0$

$\Rightarrow \quad f(n+1)=\frac{2f(2)}{n+1}$

$f(n)=\frac{1}{2n}$

$\frac{1}{f(2022)}+\frac{1}{f(2028)}=8100$

70. Let $y = y(x)$ be the solution of the differential equation $x\log_e x\frac{dy}{dx}+y=x^2\log_e x,(x>1)$. If $y(2) = 2$, then y(e) is equal to

(1) $\frac{1+e^2}{2}$ (2) $\frac{1+e^2}{4}$

(3) $\frac{4+e^2}{4}$ (4) $\frac{2+e^2}{2}$

Answer (3)

Sol. $x\ln x\frac{dy}{dx}+y=x^2\ln x$

$\frac{dy}{dx}+\frac{1}{x\ln x}\cdot y=x$

If $=e^{\int\frac{1}{x\ln x}dx}=e^{\int\frac{1}{t}dt}$, where $t=\ln x$

$=e^{\ln t}=t=\ln x$

$$y \cdot \ln x = \int x \ln x = \frac{x^2}{2}\ln x - \int \frac{x^2}{2}\cdot\frac{1}{x}$$

$$y \ln x = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C \qquad \ldots(i)$$

$$y(2) = 2 \Rightarrow C = 1$$

Putting $x = e$ in (i),

$$y = \frac{e^2}{4} + 1 = \frac{4+e^2}{4}$$

71. The shortest distance between the lines $\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}$ and $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$ is

(1) $4\sqrt{3}$ (2) $3\sqrt{3}$

(3) $5\sqrt{3}$ (4) $2\sqrt{3}$

Answer (1)

Sol. $\vec{r}_1 = \hat{i} - 8\hat{j} + 4\hat{k}$

$\vec{r}_2 = \hat{i} + 2\hat{j} + 6\hat{k}$

$\vec{a} = 2\hat{i} - 7\hat{j} + 5\hat{k}$

$\vec{b} = 2\hat{i} + \hat{j} - 3\hat{k}$

$$\text{S.D.} = \frac{\left|\vec{r}_1 - \vec{r}_2 \quad \vec{a} \quad \vec{b}\right|}{|\vec{a} \times \vec{b}|}$$

$$\left[\vec{r}_1 - \vec{r}_2 \quad \vec{a} \quad \vec{b}\right] = \begin{vmatrix} 0 & -10 & -2 \\ 2 & -7 & 5 \\ 2 & 1 & -3 \end{vmatrix}$$

$\therefore \quad 10(-16) - 2(16) = -192$

$$\left|\left[\vec{r}_1 - \vec{r}_2 \quad \vec{a} \quad \vec{b}\right]\right| = 192$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -7 & 5 \\ 2 & 1 & -3 \end{vmatrix} = 16\hat{i} + 16\hat{j} + 16\hat{k}$$

$$\vec{a} \times \vec{b} = 16\sqrt{3}$$

$$\text{S.D.} = \frac{192}{16\sqrt{3}} = 4\sqrt{3}$$

72. The letters of the word OUGHT are written in all possible ways and these words are arranged as in a dictionary, in a series. Then the serial number of the word TOUGH is

(1) 84 (2) 86

(3) 89 (4) 79

Answer (3)

Sol.

G	24
H	24
O	24
T G	6
T H	6
T O G	2
T O H	2
T O U G H	1
	89

73. The value of the integral $\int_{1/2}^{2} \frac{\tan^{-1} x}{x}\,dx$ is equal to

(1) $\frac{\pi}{4}\log_e 2$ (2) $\pi\log_e 2$

(3) $\frac{\pi}{2}\log_e 2$ (4) $\frac{1}{2}\log_e 2$

Answer (3)

Sol. $$I = \int_{1/2}^{2} \frac{\tan^{-1} x}{x}\,dx \qquad \ldots(i)$$

$$x \to \frac{1}{x}$$

$$I = \int_{1/2}^{2} \frac{1}{x}\tan^{-1}\frac{1}{x}\,dx \qquad \ldots(ii)$$

$$2I = \int_{1/2}^{2} \frac{1}{x}\cdot\frac{\pi}{2}\,dx$$

$$= \frac{\pi}{2}\ln x\Big|_{1/2}^{2} = \pi\ln 2$$

$$\Rightarrow \quad I = \frac{\pi}{2}\ln 2$$

74. Let f and g be twice differentiable functions on $\mathbb{R}$ such that

$f''(x) = g''(x) + 6x$

$f'(1) = 4g'(1) - 3 = 9$

$f(2) = 3g(2) = 12.$

Then which of the following is **NOT** true?

(1) There exists $x_0 \in (1, 3/2)$ such that $f(x_0) = g(x_0)$

(2) $|f'(x) - g'(x)| < 6 \Rightarrow -1 < x < 1$

(3) $g(-2) - f(-2) = 20$

(4) If $-1 < x < 2$, then $|f(x) - g(x)| < 8$

Answer (4)

Sol. $f''(x) = g''(x) + 6x$

$\Rightarrow f'(x) = g'(x) + 3x^2 + C$

$f'(1) = g'(1) + 3 + C$

$\Rightarrow g = 3 + 3 + C \Rightarrow C = 3$

$\Rightarrow f'(x) = g'(x) + 3x^2 + 3$

$\Rightarrow f(x) = g(x) + x^2 + 3x + C'$

$x = 2$

$f(2) = g(2) + 14 + C'$

$12 = 4 + 14 + C'$

$\Rightarrow C' = -6$

$\Rightarrow f(x) = g(2) + x^3 + 3x - 6$

$f(-2) = g(-2) - 8 - 6 - 6$

$g(-2) - f(-2) = 20$

$f'(x) - g'(x) = 3x^2 + 3$

$x \in (-1, 1)$

$3x^2 + 3 \in (0, 6)$

$\Rightarrow f'(x) - g'(x) \in (0, 6)$

$f(x) - g(x) = x^3 + 3x - 6$

At $x = -1$

$|f(-1) - g(-1)| = 10$

$\therefore$ Option (4) is false.

75. If $\vec{a} = \hat{i} + 2\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$, $\vec{c} = 7\hat{i} - 3\hat{j} + 4\hat{k}$, $\vec{r} \times \vec{b} + \vec{b} \times \vec{c} = \vec{0}$ and $\vec{r} \cdot \vec{a} = 0$. Then $\vec{r} \cdot \vec{c}$ is equal to

(1) 32 (2) 30

(3) 34 (4) 36

Answer (3)

Sol. $(\vec{r} - \vec{c}) \times \vec{b} = 0$

$\vec{r} = \lambda\vec{b} + \vec{c}$

$\vec{r} \cdot \vec{a} = 0$

$\Rightarrow \lambda\vec{b} \cdot \vec{a} + \vec{c} \cdot \vec{a} = 0$

$\Rightarrow \lambda(3) + (7 + 8) = 0$

$\Rightarrow \lambda = -5$

$\vec{r} = 5\vec{b} + \vec{c}$

$= -5\hat{i} - 5\hat{j} - 5\hat{k} + (7\hat{i} + 3\hat{j} + 4\hat{k})$

$= 2\hat{i} - 8\hat{j} - \hat{k}$

$\therefore \vec{r} \cdot \vec{c} = 17 + 24 - 4 = 34$

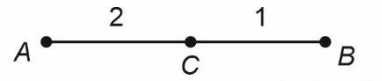
76. The plane $2x - y + z = 4$ intersects the line segment joining the points $A(a, -2, 4)$ and $B(2, b, -3)$ at the point C in the ratio 2 : 1 and the distance of the point C from the origin is $\sqrt{5}$. If $ab < 0$ and P is the point $(a - b, b, 2b - a)$ then CP^2 is equal to

(1) $\frac{73}{3}$ (2) $\frac{16}{3}$

(3) $\frac{97}{3}$ (4) $\frac{17}{3}$

Answer (4)

Sol.



$(a, -2, 4)$ $(2, b, -3)$

$c : \left(\frac{a+4}{3}, \frac{-2+2b}{3}, \frac{4-6}{3}\right)$

$= \left(\frac{a+4}{3}, \frac{-2+2b}{3}, \frac{-2}{3}\right)$

c lies on plane $2x - y + z = y$

$2\left(\frac{a+4}{3}\right) - \left(\frac{2b-2}{3}\right) - \frac{2}{3} = 4$

$\Rightarrow 2a + 8 - 2b + 2 - 2 = 12$

$\Rightarrow a - b = 2$ ...(i)

Now, $OP = \sqrt{5}$

$\left(\frac{a+4}{3}\right)^2 + \left(\frac{2b-2}{3}\right)^2 + \frac{4}{9} = 5$ and using (i)

$a = \frac{11}{5}, 1$

$\Rightarrow b = \frac{1}{5}, -1$

as also $\Rightarrow a = 1, b = -1$

$\therefore P(2, -1, -3), c\left(\frac{5}{3}, \frac{-4}{3}, \frac{-2}{3}\right)$

$CP^2 = \frac{1}{9} + \frac{1}{9} + \frac{49}{9}$

$= \frac{17}{3}$

$\therefore$ Option (4) is correct.

77. The number of 3 digit numbers, that are divisible by either 3 or 4 but not divisible by 48, is

(1) 432 (2) 507

(3) 400 (4) 472

Answer (1)

Sol. Number divisible by 3 = 300

Number divisible by 4 = 225

Number divisible by 12 = 75

Number divisible by 48 = 18

Total required number = 300 + 225 – 75 – 18

= 432

$\therefore$ Option (1) is correct.

78. The set of all values of λ for which the equation $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$ has a real solution x, is

(1) $\left[-1, -\frac{1}{2}\right]$ (2) $\left[-\frac{3}{2}, -1\right]$

(3) $\left[-2, -\frac{3}{2}\right]$ (4) $[-2, -1]$

Answer (2)

Sol. $\cos^2 2x - 2\sin^4 x - 2\cos^2 x = \lambda$

$$\cos^2 2x - \frac{1}{2}(2\sin^2 x)^2 - (1+\cos 2x) = \lambda$$

$$\Rightarrow \cos^2 2x - \frac{1}{2}(1-\cos 2x)^2 - (1+\cos 2x)$$

$$\Rightarrow \cos^2 2x - \frac{1}{2}(1+\cos^2 2x - 2\cos 2x) - 1 - \cos 2x$$

$$\Rightarrow \frac{\cos^2 2x}{2} - \frac{3}{2}$$

$\cos^2 2x \in [0, 1]$

$$\frac{\cos^2 2x}{2} \in \left[0, \frac{1}{2}\right]$$

$$\frac{\cos^2 2x}{2} - \frac{3}{2} \in \left[\frac{-3}{2}, -1\right]$$

$$\therefore \quad \lambda \in \left[\frac{-3}{2}, -1\right]$$

Option (2) is correct

79. Let $\vec{a} = 4\hat{i} + 3\hat{j}$ and $\vec{b} = 3\hat{i} - 4\hat{j} + 5\hat{k}$. If $\vec{c}$ is a vector such that $\vec{c}\cdot(\vec{a}\times\vec{b}) + 25 = 0$, $\vec{c}\cdot(\hat{i}+\hat{j}+\hat{k}) = 4$ and projection of $\vec{c}$ on $\vec{a}$ is 1, then the projection of $\vec{c}$ on $\vec{b}$ equals

(1) $\frac{1}{5}$ (2) $\frac{5}{\sqrt{2}}$

(3) $\frac{1}{\sqrt{2}}$ (4) $\frac{3}{\sqrt{2}}$

Answer (2)

Sol. $[\vec{c}\ \ \vec{a}\ \ \vec{b}] = -25$

Let $\vec{c} = l\hat{i} + n\hat{j} + n\hat{k}$

$$\begin{vmatrix} l & m & n \\ 4 & 3 & 0 \\ 3 & -4 & 5 \end{vmatrix} = -25$$

$\Rightarrow 3l - 4m - 5n = -5$ ...(i)

$\vec{c}\cdot(\hat{i}+\hat{j}+\hat{k}) = 4$

$\Rightarrow l + m + n = 4$ ...(ii)

$$\frac{\vec{c}\cdot\vec{a}}{|\vec{a}|} = 1 \Rightarrow \vec{c}\cdot\vec{a} = 5$$

$\Rightarrow 4l + 3m = 5$ ...(iii)

Using (i), (ii) and (iii)

$l = 2$, $m = -1$, $n = 3$

Now, $\frac{\vec{c}\cdot\vec{b}}{|\vec{b}|} = \frac{25}{5\sqrt{2}} = \frac{5}{\sqrt{2}}$

$\therefore$ Option (2) is correct.

80. Let R be a relation defined on N as $a\ R\ b$ if $2a + 3b$ is a multiple of 5, $a, b \in N$. Then R is

(1) an equivalence relation

(2) not reflexive

(3) symmetric but not transitive

(4) transitive but not symmetric

Answer (1)

Sol. $a\ R\ b$ if $2a + 3b = 5m$, $m \in I$

(1) $(a, a) \in R$ as $2a + 3a = 5a$, $a \in N$

Hence R is reflexive

(2) If $(a, b) \in R$ then $2a + 3b = 5m$

Now, $5(a + b) = 5n$

$3a + 2b + 2a + 3b = 5n$

$\therefore\ 3a + 2b = 5(n - m)$

$\therefore\ (b, a) \in R$

$\therefore\ R$ is symmetric

(3) If $(a, b) \in R$ and $(b, c) \in R$ then

$2a + 3b = 5m,\ 2b + 3c = 5n$

$\Rightarrow 2a + 5b + 3c = 5(m + n)$

$\Rightarrow 2a + 3c = 5(m = n - b)$

$\therefore\ (a, c) \in R$

$\therefore\ R$ is transitive

Hence R is equivalence relation.

option (1) is correct.

SECTION - B

Numerical Value Type Questions: This section contains 10 questions. In Section B, attempt any five questions out of 10. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30) using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.

81. Let $\{a_k\}$ and $\{b_k\}$, $k \in N$, be two G.P.s with common ratios r_1 and r_2 respectively such that $a_1 = b_1 = 4$ and $r_1 < r_2$. Let $c_k = a_k + b_k$, $k \in N$. If $c_2 = 5$ and $c_3 = \frac{13}{4}$ then $\sum_{k=1}^{\infty} c_k - (12a_6 + 8b_4)$ is equal to _____.

Answer (09)

Sol. $\{a_k\}$ be a G.P. with $a_1 = 4$, $r = r_1$

And

$\{b_k\}$ be G.P. with $b_1 = 4$, $r = r_2$ $\quad (r_1 < r_2)$

Now

$C_k = a_k + b_k$

$c_1 = 4 + 4 = 8$ and $c_2 = 5$

$a_2 + b_2 = 5$

$\therefore\ r_1 + r_2 = \frac{5}{4}$

and $c_3 = \frac{13}{4} \Rightarrow r_4^2 + r_2^2 = \frac{13}{16}$

$\therefore\ \frac{25}{16} - 2r_1r_2 = \frac{13}{16} \Rightarrow 2r_1r_2 = \frac{3}{4}$

$\therefore\ r_2 - r_1 = \sqrt{\frac{25}{16} - \frac{3}{2}} = \frac{1}{4}$

$\therefore\ r_2 = \frac{3}{4},\ r_1 = \frac{1}{2}$

$\therefore\ a_6 = 4 \times \frac{1}{2^5} = \frac{1}{8},\ b_4 = 4 \times \frac{27}{64} = \frac{27}{16}$

and $\sum_{K=1}^{\infty} C_K = 4\left[\frac{1}{1-\frac{1}{2}} + \frac{1}{1-\frac{3}{4}}\right] = 24$

$\therefore\ \sum_{K=1}^{\infty} C_K - (12a_6 + 8b_4) = 09$

82. the total number of 4-digit numbers whose greatest common divisor with 54 is 2, is _____.

Answer (3000)

Sol. $gcd(a, 54) = 2$ when a is a

4 digit no.

And $54 = 3 \times 3 \times 3 \times 2$

So, a = all even no. of 4 digits

– Even multiple of 3 (4 digits)

= 4500 – 1500

= 3000

83. If the equation of the normal to the curve $y = \frac{x - a}{(x + b)(x - 2)}$ at the point (1, –3) is $x - 4y = 13$, then the value of $a + b$ is equal to _____.

Answer (04)

Sol. Given curve : $y = \frac{x - a}{(x + b)(x - 2)}$ at $(1, -3)$

$\therefore\ -3 = \frac{1 - a}{(1 + b)(-1)} \Rightarrow 3 + 3b = 1 - a$

$\beta \Rightarrow a + 3b + 2 = 0$

$y = \frac{x - a}{(x + b)(x - 2)}$

$\frac{dy}{dx} = \frac{(x + b)(x - 2) - (x - a)[(x + b) + (x - 2)]}{[(x + b)(x - 2)]^2}$

at (1, –3) $m_T = \frac{-(1 + b) - (1 - a)(b)}{(1 + b)^2} = -4$

$\therefore \quad 1 + b + b - ab = 4(1 + b)^2$

$\Rightarrow \quad 1 + 2b + b(3b + 2) = 4b^2 + 4 + 8b$

$\Rightarrow \quad b^2 + 4b + 3 = 0$

$(b + 1)(b + 3) = 0$

$b = -1, a = 1$ but $1 + b \neq 0$

$b = -3, a = 7$ $\therefore \quad b \neq -1$

$\therefore \quad a + b = 04$

84. A triangle is formed by the tangents at the point (2, 2) on the curves $y^2 = 2x$ and $x^2 + y^2 = 4x$, and the line $x + y + 2 = 0$. If r is the radius of its circumcircle, then r^2 is equal to _____.

Answer (10)

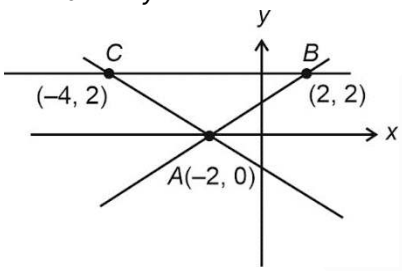
Sol. Tangent for $y^2 = 2x$ at (2, 2) is

$L_1 : 2y = x + 2$

Tangent for $x^2 + y^2 = 4x$ at (2, 2) is

$L_2 : y = 2$

$L_3 : x + y = 2 = 0$



Radius of circumcircle $= \dfrac{abc}{4\Delta}$

$$= \frac{\left(\sqrt{20}\right)(6)\left(\sqrt{8}\right)}{4 \times \frac{1}{2} \times 6 \times 2}$$

$$R = \sqrt{10}$$

$R^2 = 10$

85. Let $X = \{11, 12, 13, \ldots\ldots, 40, 41\}$ and $Y = \{61, 62, 63, \ldots\ldots, 90, 91\}$ be the two sets of observations. If $\bar{x}$ and $\bar{y}$ are their respective means and σ^2 is the variance of all the observations in $X \cup Y$, then $\left|\bar{x} + \bar{y} - \sigma^2\right|$ is equal to _____.

Answer (603)

Sol. $x = \{11, 12, 13 \ldots, 40, 41\}$

$y = \{61, 62, 63, \ldots, 90, 91\}$

$$\bar{x} = \frac{\frac{31}{2}(11 + 41)}{31} = \frac{1}{2} \times 52 = 26$$

$$\bar{y} = \frac{\frac{31}{2}(61 + 91)}{31} = \frac{1}{2} \times 152 = 76$$

$$\sigma^2 = \frac{\Sigma x_i^2 + \Sigma y_i^2}{62} - \left(\frac{\Sigma x + \Sigma y}{62}\right)^2$$

$= 705$

Now

$\left|\bar{x} + \bar{y} - \sigma^2\right|$

$= |26 + 76 - 705|$

$= 603$

86. Let $\alpha = 8 - 14t$, $A = \left\{z \in \mathbb{C} : \dfrac{\alpha z - \bar{\alpha}\bar{z}}{z^2 - (\bar{z})^2 - 112i}\right\}$ and $B = \{z \in \mathbb{C} : |z + 3i| = 4\}$. Then $\sum\limits_{z \in A \cap B} (\text{Re}\, z - \text{Im}\, z)$ is equal to _____.

Answer (07)

Sol. Let $z = x + iy$

and $\alpha = 8 - 14\, i$

$$\frac{\alpha z - \bar{\alpha}\,\bar{z}}{z^2 - \bar{z}^2 - 112i} = 1$$

$$\therefore \quad \frac{(16y - 28x)i}{4xy - 112i} = 1$$

$(16y - 28x + 112)i = 4xy$

$\therefore \quad z = -7i$ or 4

Now, $z = -7i$ satisfy B

$B : x^2 + (y + 3)^2 = 16$

$A \cap B = (0, -7)$

$Re\, z - Im\, z = 7$

87. Let A be a symmetric matrix such that $|A| = 2$ and $\begin{bmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{bmatrix} A - \begin{bmatrix} 1 & 2 \\ \alpha & \beta \end{bmatrix}$. If the sum of the diagonal elements of A is s, then $\dfrac{\beta s}{\alpha^2}$ is equal to _____.

Answer (05)

Sol. $A = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$

$|A| = ab - c^2 = 2 \qquad \ldots(1)$

$$\begin{pmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{pmatrix} \begin{pmatrix} a & c \\ c & b \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ \alpha & \beta \end{pmatrix}$$

$2a + c = 1$ ...(2)

$2c + b = 2$ ...(3)

$3a + \frac{3}{2}c = \alpha$ ...(4)

$3c + \frac{3}{2}b = \beta$ ...(5)

From (1), (2) and (3)

$a = \frac{3}{4}\quad b = 3\quad c = -\frac{1}{2}$

$\Rightarrow$ Now, $\alpha = \frac{6}{4}$

$\beta = 3$

$s = \frac{15}{4}$

$$\frac{\beta s}{\alpha^2} = \frac{3 \times \frac{15}{4}}{\left(\frac{6}{4}\right)^2} = \frac{\frac{45}{4}}{\frac{9}{4}} = 5$$

88. Let $a_1 = b_1 = 1$ and $a_n = a_{n-1} + (n-1)$, $b_n = b_{n-1} + a_{n-1}$, $\forall n \geq 2$. If $S = \sum_{n=1}^{10} \frac{b_n}{2^n}$ and $T = \sum_{n=1}^{8} \frac{n}{2^{n-1}}$, then $2^7(2S - T)$ is equal to _____.

Answer (461)

Sol. $\because a_n = a_{n-1} + (n-1)$ and $a_1 = b_1 = 1$

$b_n = b_{n-1} + a_{n-1}$

$\therefore \quad b_{n+1} = 2b_n - b_{n-1} + n - 1$

n	b_n	$b_n - n$
1	1	0
2	2	0
3	4	1
4	8	4
5	15	10
6	26	20
7	42	35
8	64	56
9	93	84
10	130	120

$$\therefore \quad 2S - T = \left(\sum_{n=1}^{8} \frac{b_n - n}{2^{n-1}}\right) + \frac{b_9}{2^8} + \frac{b_{10}}{2^9}$$

$$= \frac{461}{128}$$

$\therefore \quad 2^7(2S - T) = 461$

89. A circle with centre (2, 3) and radius 4 intersects the line $x + y = 3$ at the points P and Q. If the tangents at P and Q intersect at the point $S(\alpha, \beta)$, then $4\alpha - 7\beta$ is equal to _____.

Answer (11)

Sol. The line $x + y = 3$ ...(i)

is polar of $S(\alpha, \beta)$ w.r.t. circle

$(x-2)^2 + (y-3)^2 = 16$

$\Rightarrow \quad x^2 + y^2 - 4x - 6y - 3 = 0$

Equation of polar is

$\alpha x + \beta y - 2(x + \alpha) - 3(4 + \beta) - 3 = 0$

$(\alpha - 2)x + (\beta - 3)y - (2\alpha + 3\beta + 3) = 0$ ...(ii)

(i) and (ii) represent the same.

$$\therefore \quad \frac{\alpha - 2}{1} = \frac{\beta - 3}{1} = \frac{2\alpha + 3\beta + 3}{3}$$

$\alpha - \beta + 1 = 0$

$\alpha - 3\beta - 9 = 0$

$\Rightarrow \quad \alpha = -6, \beta = -5$

$4\alpha - 7\beta = 11$

90. Let $\alpha_1, \alpha_2, ..., \alpha_7$ be the roots of the equation $x^7 + 3x^5 - 13x^3 - 15x = 0$ and $|\alpha_1| \geq |\alpha_2| \geq ... \geq |\alpha_7|$. Then $\alpha_1\alpha_2 - \alpha_3\alpha_4 + \alpha_5\alpha_6$ is equal to _____.

Answer (09)

Sol. $x^7 + 3x^5 - 13x^3 - 15x = 0$

$x(x^6 + 3x^4 - 13x^2 - 15) = 0$

$x = 0 \quad = \alpha_7$

Let $x^2 = t$

$t^3 + 3t^2 - 13t - 15 = 0$

$(t + 1)\ (t + 5)\ (t - 3) = 0$

$t = x^2 = -1, -5, 3$

$x = \pm i, \pm\sqrt{5}i, \pm\sqrt{3}$

$\alpha_1, \alpha_2 = \pm\sqrt{5}i, \alpha_3, \alpha_4 \pm = \sqrt{3}, \alpha_5, \alpha_6 = \pm i$

$\alpha_1\alpha_2 - \alpha_3\ \alpha_4 + \alpha_5\ \alpha_6 = 5 + 3 + 1 = 9$

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